

Welded And Adhesively Bonded Panel Replacement (EXT02)

Textbook



Version 4.2

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EXT02-STMAN1-E

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Introduction

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Obligations To The Customer And Liability



The collision repair industry has an obligation to correctly repair the customer's vehicle. Collision repairs must be performed using:

- recommended or tested procedures from vehicle makers, I-CAR, and other research and testing organizations.
- quality replacement parts and materials.
- repair processes and parts as written and agreed upon in the repair order. If items on the repair agreement are not consistent with the repair order, it can be considered fraud.

Performing proper collision repairs requires using parts and procedures that keep remaining warranties intact.

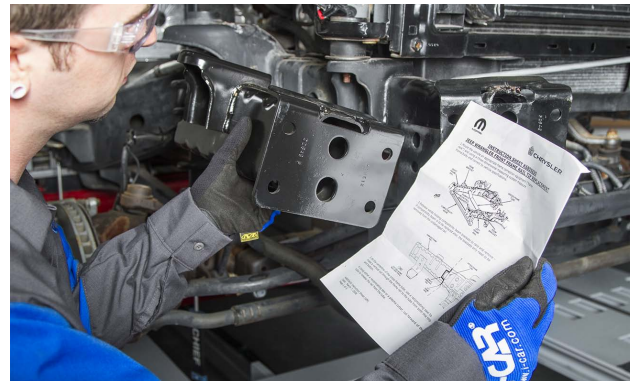
Collision repairs must restore:

- safety.
- structural integrity.
- durability.
- performance.

- fit.
- finish.

Throughout the damage analysis and repair process the repairer and insurer must:

- communicate with each other.
- maintain constant communication with the customer.
- be in agreement with each other and the customer on how repairs will be performed.
- inform the customer of any changes in the repair plan from the original repair agreement, and explain the changes and why they have to be made.



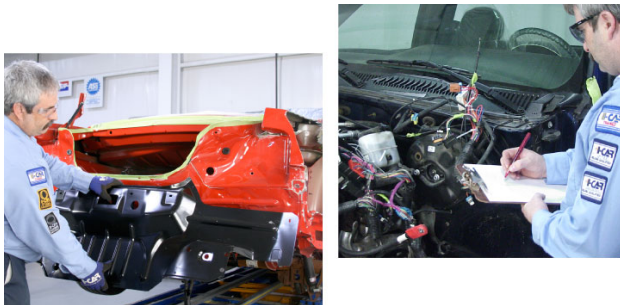
To reduce liability:

- make sure that all repairs are performed thoroughly, correctly and as listed in the damage report.
- follow proper procedures.
- have documentation of required repairs with detailed record keeping available for customers.

Technicians are considered the experts and are expected to be knowledgeable on how to perform a quality repair.

Liability insurance that covers the repair facility may not always cover all damages. For example:

- the policy may not cover faulty repairs, leaving liability responsibility completely on the facility.
- a shop owner may find that repair facility liability coverage may not cover the full amount awarded in a lawsuit. The shop owner would have to pay the difference.

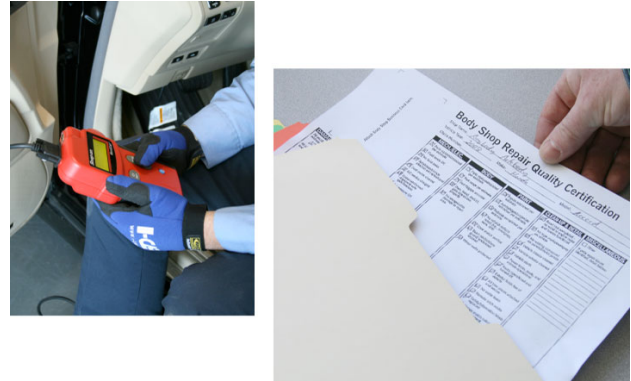


It is difficult to reduce the risk of liability exposure. The part that the repairer can control is the chance of being found at fault. Chances can be minimized by:

- using recommended or tested procedures from the vehicle makers, I-CAR, or other research and testing organizations.
- using quality replacement parts and materials that restore fit,

finish, durability, and perform at least as well as the original.

- keeping thorough records.



Keeping thorough records includes more than recording the date, mileage, and pre-existing damage. Record keeping also includes:

- making sure all notes are legible.
- verifying the repairs that were made or not made.
- having the customer sign a waiver for repairs that they do not want performed. Repairers must determine their liability on not repairing safety systems such as restraint and anti-lock brake systems.
- keeping computer printouts or worksheets on file showing wheel alignment readings or vehicle dimensions before and after repairs.
- keeping scan tool printouts and records of computer codes for airbag, anti-lock brake, emission, and powertrain control module (PCM) systems.

- attaching the OEM or other tested procedure printout to the vehicle repair order.
- keeping receipts for all sublet work performed.



Refer to "Video: Topics Off Limits" in the presentation. This video identifies topics that should not be brought up in class.

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Module 1 - Exterior Panel Replacement

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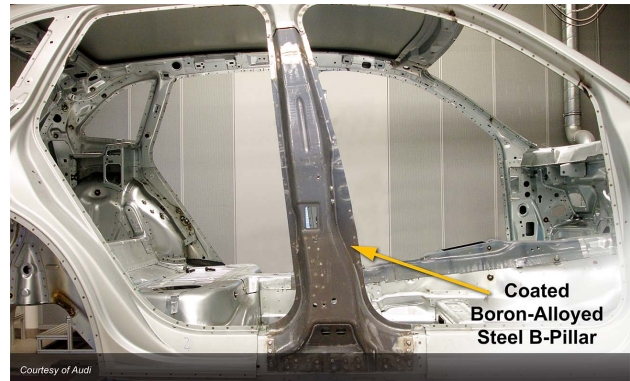
Panel Construction

Learning objectives for this module include:

- identifying types of panel construction.
- analyzing and measuring for damage.
- considerations for making repair or replace decisions.
- determining what to include in a repair plan.
- understanding the function of silicon bronze and brazing.
- removing and installing spot welds and GMA (MIG) welds.



This bumper cover and rear trim piece are made of plastic.



The B-pillar on this Audi Q5 uses ultra high strength steel.



This 2013 Lincoln MKT Wagon uses aluminum for the outer liftgate panel.



The roof panel on this 2009 Chevrolet Corvette ZR1 is made of a Carbon Fiber composite.

Exterior panels are made of:

- plastic.
- steel.
- aluminum.

- composites.

Common plastic panels include many bumper fascias and grille assemblies.

Mild steel is the most common type of material found in vehicle exterior panel construction. It may also be used in structural parts and frame assemblies. Harder steels, such as ultra-high-strength steel (UHSS) and high-strength steel (HSS), are commonly used in bumper reinforcements, door intrusion beams, and some structural parts. HSS can also be used on exterior body panels.

One type of ultra-high-strength steel that is often used in vehicle construction is boron-alloyed steel. Using this material, panels can be made thinner to reduce the weight because the steel is so strong. One characteristic of boron-alloyed steel is that it's very hard, making it difficult to metal work or drill. Some repair considerations when working with boron-alloyed steel include:

- if bent, it cannot be straightened. It must be replaced.
- it rapidly weakens if heated.
- it may be GMA (MIG) welded, but caution must be taken due to weakening in the heat-affect zone.
- being difficult to cut or drill. Special blades or bits are available.

Aluminum is commonly being used for individual panels or the entire vehicle. For example, aluminum is used throughout

the unitized structure on the 2015 Ford F-150.

Composite use is ever increasing with vehicle makers looking to reduce vehicle weight while increasing strength. Composites include carbon fiber, sheet-molded compound (SMC), and fiberglass.



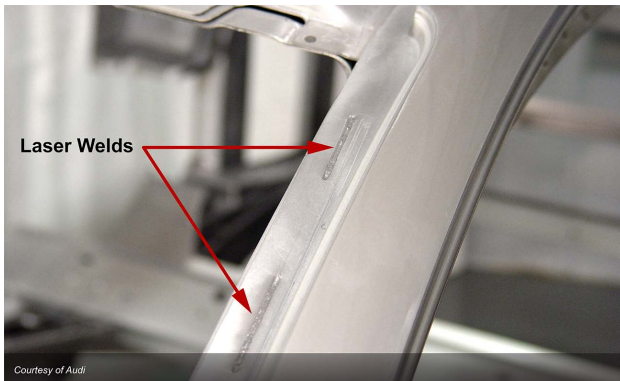
These are examples of spot welds used for panel attachment.



Exterior bolted panels may require fastener replacement.



Some vehicle maker will recommend an adhesive for panel replacement when bolts, rivets, or a rope hem flanges are used.



Laser welds are becoming more common in today's vehicles.



Riveted exterior panels may require adhesive for panel replacement.



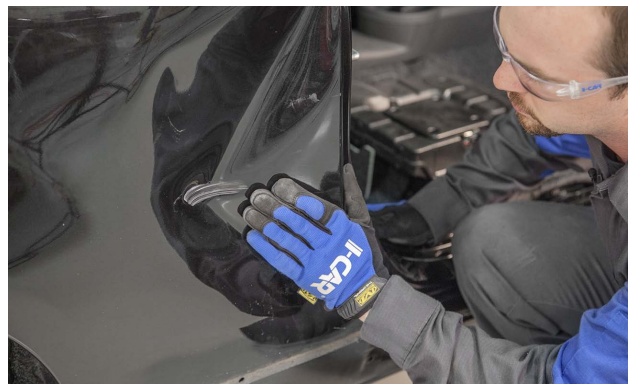
Hem flanges are commonly found around door skins and wheel wells.

Panels are attached to the vehicle structure by one or a combination of:

- spot welds.
- mechanical fasteners.
- adhesive.
- weld bonding.
- rivets.
- hem flanges.

The type of attachment method used may depend on the material type and panel configuration. For example, plastic panels are either mechanically fastened or bonded to the vehicle structure.

Damage Analysis



This technician has discovered damage to the quarter panel during a visual analysis.



Misalignment of door gaps is an indication of damage.



Cracked or damaged welds is an indication of possible structural damage.

To determine the extent of panel damage, perform a visual analysis and make measurements.

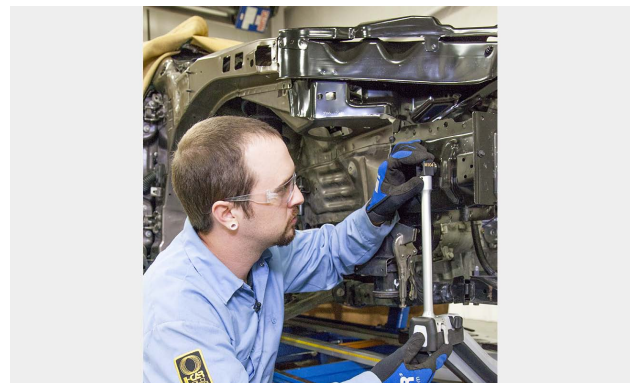
Visual analysis requires looking for:

- exterior panel damage such as kinks, dents, and dings.
- panel misalignment. An indication of misalignment may be gaps that are not uniform.
- broken, damaged or separation of attachment points. This may include welded panel installations or rivet bonded.

Panel damage may cause air and water leaks in the passenger compartment. This may result in corrosion, damage to the vehicle interior, and excessive wind noise.

Vehicle Protection

Cover all areas that may allow moisture to enter into the vehicle interior. These areas may include broken windows, bent door frames, etc.



This technician is making three-dimensional measurements to determine if there is structural damage.

Measure the vehicle structure and surrounding area. Determine if the vehicle structure or the panel requires straightening. If the structure requires straightening, the vehicle must be mounted to a frame rack or bench and pulled to the proper dimensions. The vehicle panel may be left in place during the structural straightening process to help direct the pull.



This technician is measuring for damage using a tram bar in the engine compartment.

Additional measurements used to check for damage include measuring and verifying the dimensions of door openings, window openings, tailgate openings, etc. Since some dimension guides provide measurements for the structure only, it may be necessary to use an alternate source. If no vehicle specifications exist, make comparative measurements from the left side of the vehicle to the right side. A vehicle that is the same make and model can also be used to make comparative measurements.



This 2007 Jeep Wrangler is being straightened before panel replacement to ensure proper alignment of all panels.

Measuring and straightening vehicle damage before repairing the panel ensures that the:

- panel fits properly when installed.
- openings are the proper dimensions.



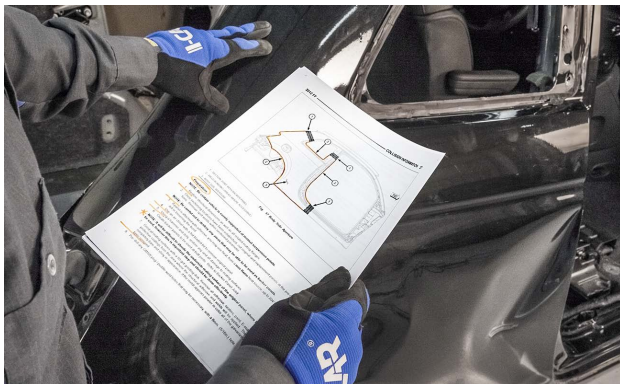
There are many considerations when determining if the panels damaged on this Audi Q5 should be repaired or replaced.

Determining repairability requires considering:

- cost of time and materials. A panel that requires considerable time and materials to repair is generally not as cost effective as replacing the panel.
- which type of repair is least intrusive. Examples of an intrusive repair may include quarter panel replacement with the removal of many vehicle maker spot welds, striker, backlite, wheelhouse, etc. Repair may be a better alternative than replacing the entire panel. Also, it is possible that the replacement of the panel may

- require removal of an adjacent panel.
- durability of the repair.
- the amount of vehicle maker corrosion protection that must be removed. When removing spot welds to replace a quarter panel, more vehicle maker corrosion protection will be removed than if the panel were left in place.
- if the damage requires accessing inner panel damage.
- if the part is available.
- if the part is sold as an assembly.

Planning The Repair



When developing a repair plan, the vehicle maker's procedure is useful for making panel sectioning decisions.

When developing a repair plan, determine:

- how many panels must be replaced following damage analysis guidelines.
- what, if anything, must be disassembled. This includes exterior trim panels, interior trim panels, wiring, airbag parts, etc.
- panel sectioning decisions. Sectioning decisions will be based largely on vehicle maker repair recommendations and general guidelines.
- attachment method(s) (squeeze-type resistance spot welding (STRSW), GMA (MIG) plugs welds, adhesives, rivet bonding).
- the sequence for replacing the panels.

Sequence plays an important role in multiple panel replacement. It is recommended to install replacement panels starting with the panel adjacent to an undamaged panel. This helps ensure proper panel gaps for all replacement panels.



The quarter panel flange around the door opening on this vehicle is held together with spots welds.

Replacing at factory seams may:

- be recommended by the vehicle maker.
- be based on part design and size.
- require replacement of adjacent panels. For example, a flange of the adjacent panel may be over the top of the damaged panel.

Therefore, the adjacent panel flange must be removed.

- require removing spot welds, adhesive, or both around the entire panel.



This sectioning procedure indicates that the undamaged roof panel does not need to be removed.

Sectioning a panel may:

- be performed to limit the amount of intrusion caused by the replacement of an entire panel.
- may eliminate the need to remove an undamaged adjacent panel.

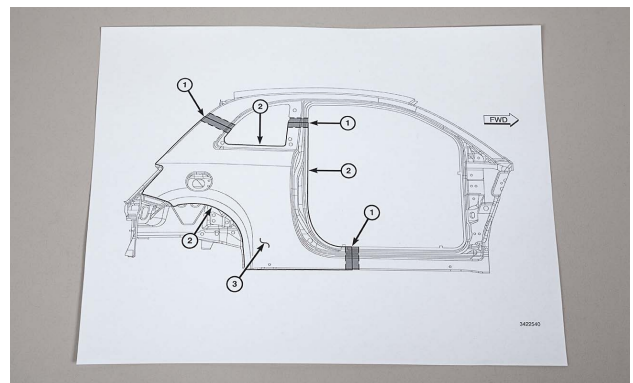
Part design may play a factor in determining sectioning locations.

Some vehicle maker sectioning procedures allow a panel to be replaced while leaving the glass in place.



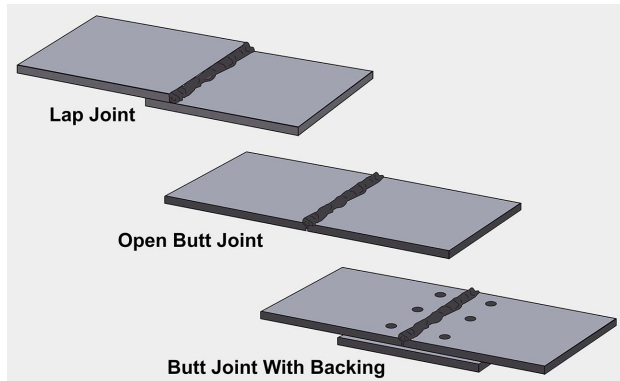
This replacement unside includes the A-pillar, B-pillar, rocker panel, and the quarter panel.

A unside panel is an exterior one-piece panel that typically contains the B-pillar, rocker panel, and A-pillar. Some unsides contain the quarter panel, including the sail area.



The Fiat 500 has sectioning procedures for the replacement quarter panel.

It may not be necessary to install the entire unside when replacing a damaged area. Many vehicle makers provide recommended sectioning locations. Sometimes a replacement panel, such as a unside, may arrive initially cut for the panel being replaced.

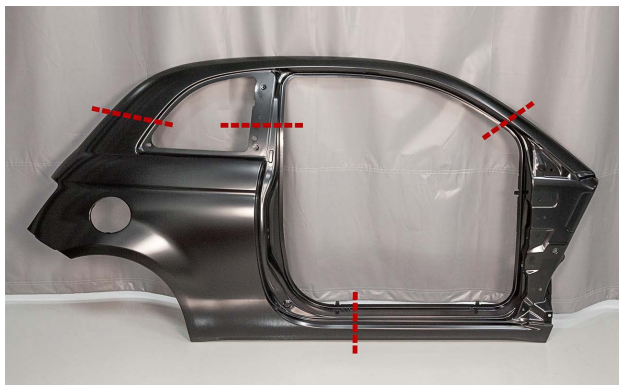


The type of joint used is typically recommended by the vehicle maker.

Types of sectioning joints include:

- lap joint.
- open butt joint.
- butt joint with backing.

The joint used may depend on vehicle maker recommendations. If a backing is used, some vehicle makers may also require plug welds into the backing.



This image shows some common sectioning locations for a uniside.

Common sectioning locations include the:

- sail panel and belt line.

- A-pillar.
- door skin at the belt line.
- dog leg / rocker area.

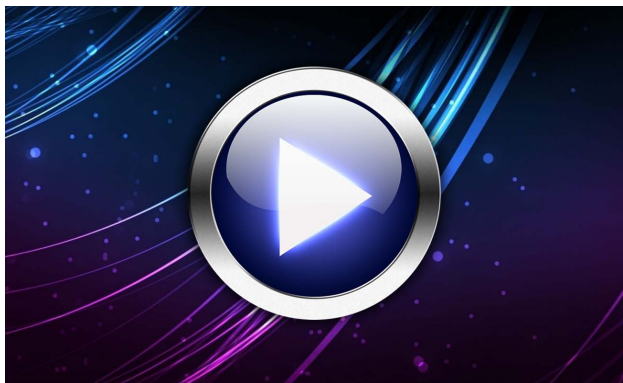
Sectioning joint cut methods include “cut and join” and “lap and scribe.”



Placing a piece of scrap stock between the panel and reinforcement will help prevent damaging the reinforcement while cutting.

Reinforcements should not be cut when cutting a part for sectioning. Cutting into a reinforcement generally requires the reinforcement to be replaced. Therefore, it is important to identify what is behind panels before cutting operations begin. Reinforcements may be identified:

- in the body repair manual.
- by panel flanges. Panel flanges with three or more visible edges has an interior panel.
- by spot weld configuration. Some spot welds that appear in the middle of a panel are a good indication that a reinforcement is present.



Refer to Video: Cutting To Avoid Reinforcements in the presentation. This video demonstrates how to identify and avoid reinforcements when cutting outer body panels.



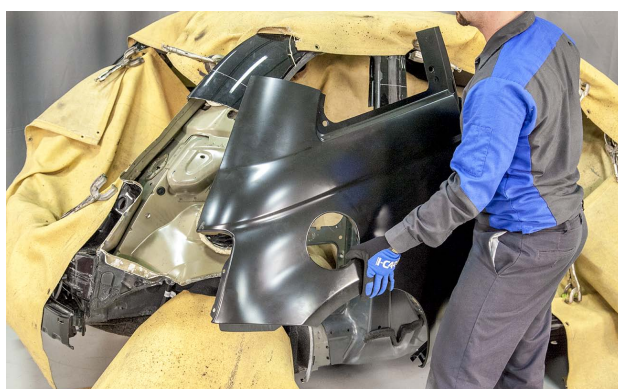
Measurements are marked on the replacement part so that the cut will overlap the original part.



This technician is using a saw to cut the replacement part for a lap and scribe sectioning joint.



The replacement part overlaps the vehicle part and the technician scribes a mark.



With the lap and scribe marks made, the technician removes the replacement panel in preparation for the final cuts on the vehicle.



A grinding wheel is used to make a clean cut along the scribe mark.

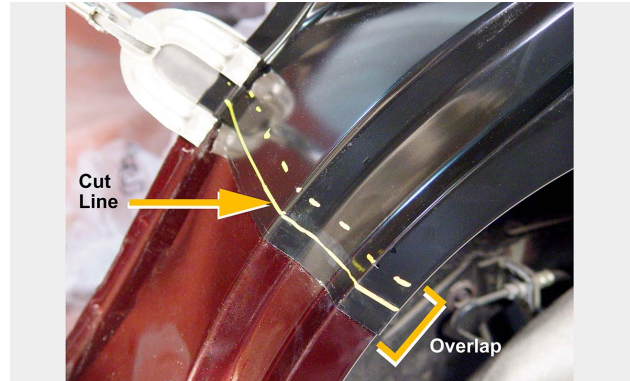
To perform a "lap-and-scribe" sectioning cut method:

- make measurements and perform an initial cut on the original part on the vehicle and the replacement part. Make measurements so that the replacement part will slightly overlap the original part.
- clamp the replacement part in place. The replacement part will slightly overlap the original part.
- make a scribe mark where the replacement part overlaps the original part.
- remove the replacement part.
- make a cut next to the scribe mark.

At this point, the joint should be very close to matching between the replacement part and original part. Grind back one of the panels to match up for an even fit.



Refer to Video: Lap-And-Scribe in the presentation. This video discusses concepts on how to perform a lap-and-scribe sectioning joint.



The technician has marked a cut line in the middle of the overlap area.

To perform a “cut-and-join” sectioning cut method:

- make measurements and make an initial cut on the original part and replacement part. Similar to the “lap-and-scribe,” make measurements so that the replacement part will slightly overlap the original part.
- install the replacement part.
- make a cut in the middle of the overlap area. This will simultaneously cut both the replacement part and the original part. The result is a joint line between both panels that exactly match.
- remove remaining scrap material.



Refer to Video: Cut-And-Join in the presentation. This video discusses concepts on how to preform a cut-and-join sectioning joint.



This is an example of a hem flange formed by the vehicle maker.

Exterior panels that are hemmed include:

- hoods and rear closure panels. The inner and outer panels are hemmed together and replaced as assemblies. Hoods and rear closure panels are only hemmed by the vehicle maker.
- quarter panels. Quarter panels may require hems along the wheelhouse flange.

- door panels. Door panels will require hems along the perimeter of the replacement panel.

Hem flanges are formed by folding the edge of the outer panel over itself and the inner panel. This clamps the inner panel into the fold. Before the panel is installed and the edge is clamped, adhesives are used in the folded area to seal the flange.

General Panel Removal Procedures



This vehicle is being prepped for a quarter panel replacement.

Initial steps for exterior panel replacement may include:

- disconnecting required wiring.
- removing bolted-on parts (cladding, trim, mirrors, emblems, etc.).
- removing glass, such as the windshield and backglass.
- removing seats and center consoles.
- straightening the vehicle to proper dimensions.

Be sure to store fasteners so that they can be easily matched to the right part and location when reinstalling.



The C-clamp style drill can be used when both sides of the flange are exposed.



After drilling out spot welds, a chisel and hammer may be needed to release the flange.



An air powered chisel is effective removing spot welds when a mild steel outer panel is welded to a high-strength steel reinforcement.



A belt grinder is an effective tool for removing spot welds.

The method of removing the spot welds, and the tool used, depends on:

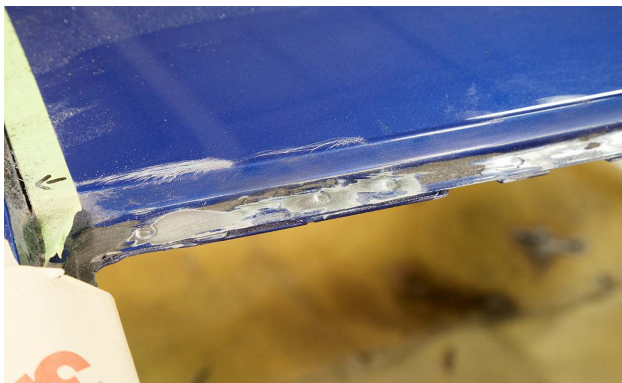
- access to the spot welds.
- if one part will be reused. Some methods, such as grinding or using a cutoff wheel, make the removed part unusable.
- the side that the parts will be welded from. Some methods, like drilling all the way through both panels (commonly done when using two twist drill bits), require plug welds be made from the backside. Drilling through both panels eliminates the need to punch plug weld holes into the replacement panel. Make sure there is access on the opposite side to weld the panel.



Refer to Video: Spot Weld Removal Tools in the presentation. This video discusses the different ways and the tools used to remove spot welds.



A wire brush grinding wheel is used to remove coatings from the windshield pinchweld.



With the coatings removed, the spot welds are exposed.



This technician is using a c-clamp style drill to remove the spot welds, taking care to only drill through the top layer of the flange.

Removing parts at factory seams begins by first locating, then removing the vehicle maker original spot welds on the flanges. Locate the spot welds by:

- removing paint, seam sealers, and undercoating until spot welds are visible. An outline of the spot welds will appear.
- using a body repair manual to locate reinforcements and hidden spot welds.

Depending on the tool, a center punch may be useful to prevent the drill bit from wandering. When removing spot welds, make sure not to drill through the inner panel flange if welding is to be performed on the outside of the panel.

These tools are sensitive to heat and require cooling between cuts. Make sure that they are properly sharpened or replaced when dull.



An air chisel is used to separate the mild steel panel from the boron-alloyed steel panel.

Some parts are made of boron-alloyed steel, a very high-strength steel that is difficult to drill. To remove these parts using a drill, generally it helps to drill the spot welds from the opposite side, if the steel is a lower-strength steel.

In many cases, an air chisel can be used to separate the mild steel panel from the very high-strength steel by working the chisel between the two metals and breaking the spot weld free.

Caution must be taken to prevent damaging the boron alloyed steel reinforcement. If damaged, the reinforcement will need to be replaced. Boron-alloyed steel or any ultra-high-strength steel cannot be repaired.



After the spot welds were drilled, this technician is using a hammer and chisel to complete the spot weld removal on this quarter panel.

Cutting tools may also be used to cut access holes in panels to assist in the panel removal process. This is sometimes done to allow access to the flange area when separating bonded panels.

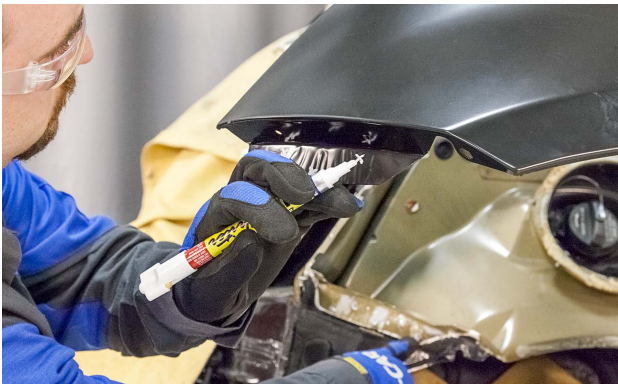
Once the spot welds have been drilled, a hammer and chisel can be used to separate the two panels.

At this point, think about the attachment method to help determine appropriate flange treatment.

After the spot welds are removed, use a cutoff wheel or grinder to remove remaining nuggets. Be sure not to damage the:

- remaining flange.
- E-coat.

If there are any residual noise, vibration, and harshness (NVH) materials, heat may be required to loosen these materials to remove the panel.



The technician is marking the plug weld locations on the replacement part using the same locations as the original welds.

After removing the panel spot welds, mark or scribe the same quantity and location of replacement welds on the replacement panel. If using STRSW, however, do not use the same locations for replacement spot welds. Once an area has been spot welded, the metal in that area has thinned. Using the same area may result in a weak weld.

To transfer vehicle maker spot weld locations for GMA (MIG) plug welds, transfer spot weld measurements from the damaged panel to the replacement panel. A second method, if the panel design allows, is to hold the replacement panel against the damaged panel and transfer the location of the holes on the damaged panel to the holes on the undamaged panel.



An infrared heater is being used to loosen the adhesive bond on this door skin.

To remove a bonded panel using heat:

- cut access holes in the panel if required.
- grind the outer edge of the panel (a procedure used for hem flanges only).
- heat the adhesive using a heat gun or infrared lamp.
- use a chisel to pry apart the two panels.

After the panel is removed, use a wire wheel or scraper to remove any loose or residual adhesive from the bond area.



A pneumatic knife is used to cut the adhesive bond on this roof panel.

To remove a bonded panel using mechanical cutters:

- determine if a layer of existing adhesive needs to be left intact.
- remove any mechanical fasteners.
- use a piano wire or pneumatic knife to cut the adhesive.
- remove loose urethane adhesive when done.
- trim the remaining urethane adhesive to remove any high spots. High spots may result in the uneven application of replacement adhesives.

The type of flange design may dictate the type of method used to separate the adhesive bond. For example, a roof structure design may prevent a pneumatic knife from accessing the adhesive. Therefore, piano wire may be a better tool for cutting the urethane adhesive. Examples of vehicles that use a one-part urethane to bond a roof panel to the vehicle structure include the 2012 Chevrolet Malibu and the 2013 Cadillac CTS.

Personal Safety

Personal protection equipment includes:

- hearing protection.
- a NIOSH-approved particulate respirator.



A hem flange is being removed with a hem flange pry tool (left) and an angle grinder (right).

When removing an adhesively bonded panel along a hemmed area:

- use a prying tool that is inserted along the edge to separate the panels.
- separate the panels by applying heat to soften the adhesive.
- grind the outer edge of the panel using a 36-grit abrasive.

When separating panels, use care to not damage the interior mating flange.

General Panel Preparation Methods



This technician is using a plastic-coated abrasive wheel to remove adhesive on this mating surface.



A hammer and dolly is used to straighten the mating flanges.

Preparing a mating flange after outer panel removal requires:

- cleaning the flange so it is free of corrosion and / or adhesive. While cleaning the flange, try to avoid removing the zinc coating. A plastic coated abrasive wheel may be used to clean a flange of remaining adhesive.
- straightening mating panel flanges.
- applying corrosion protection to bare steel areas along the flange. The type of corrosion protection will vary based on the method of attachment. After applying corrosion protection, it may be beneficial to remove primer / E-coat from the weld zones to improve weld quality.



Both sides of this replacement part coatings are removed in preparation for weld bonding.

Preparing a replacement panel requires:

- straightening mating flange deformations. This is typically done using a hammer and dolly.
- removing primer from the weld zone.
- cleaning the mating flange.
- applying corrosion protection to bare steel areas. The type of corrosion protection will vary based on the method of attachment.

General Panel Attachment Methods



Locking pliers are used to temporarily attach the panel while test-fitting.

Test-fitting a panel:

- is done to check the alignment with adjacent panels.
- requires Locking pliers to hold the panel in place.

Screws may be used in place of locking pliers to hold the panel in place. Screws may also be used to help realign the replacement panel if it must be removed after fit testing.



This technician is using a panel gap measuring tool.

Some vehicle makers provide panel fit standards to:

- prevent excessive wind noise and water leaks.
- ensure a uniform gap between panels.

Examples of vehicle makers that provide gap widths and flushness heights include:

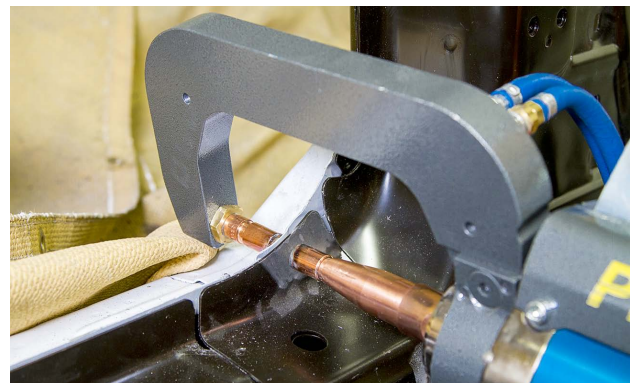
- Audi.
- Toyota.

- Lexus.
- Chrysler.

Common panel gaps are between 3 - 8 mm for most vehicles.

If no specifications exist from the vehicle maker, make sure that the gaps are even around the entire panel or correspond with the panel gaps on the opposite side of the vehicle.

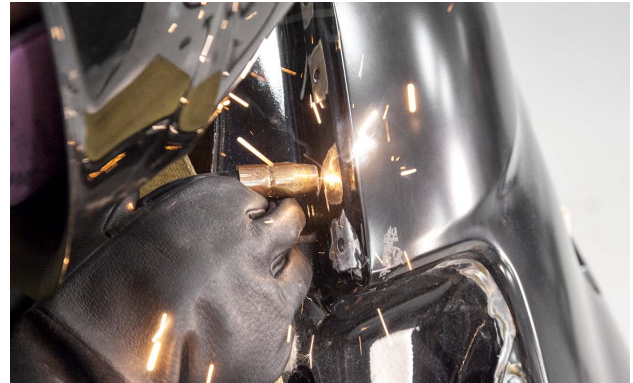
Check panel fit with the suspension loaded as the body may flex with the suspension unloaded. This may cause gaps to be wide at one point and narrow at another. If the panels are adjusted while the vehicle is flexed, once the suspension is loaded again, the panel gaps will change. It is also possible that weld-on panels will dent or buckle once the suspension is loaded.



This exterior panel flange has both sides exposed, making the spot welder a good choice for attachment.



This technician is using plug welds to attach a quarter panel.



When welding, alternate plug weld locations to control heat / panel warpage.

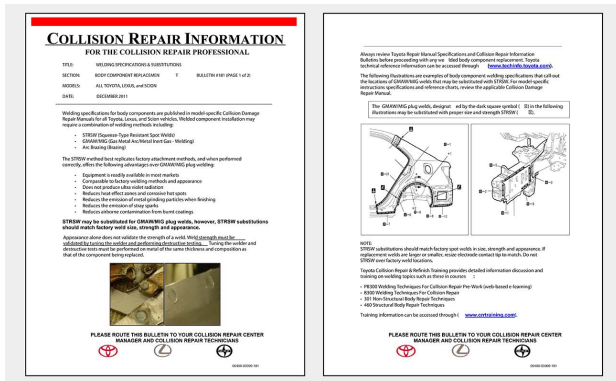
After test fitting the panel, proper welding methods must be chosen. Welding methods include:

- STRSW.
- MIG Brazing.
- GMA (MIG) welding.
- GMA (MAG) welding.
- a combination of STRSW and GMA (MIG) plug welds.

Follow vehicle maker recommendations regarding the type of welding method to use. If no welding recommendations exist from the vehicle maker, the repairer may decide on the best welding method for exterior panels.

When installing welded-on panels:

- perform a final fit test by installing the panel and securing with temporary fasteners. Check the panel alignment with adjacent panels. Remove the panel.
- punch or drill holes in the replacement panel flanges (if required).
- apply weld-through primer to bare metal areas if recommended by the vehicle maker.
- make test welds on steel panels that are the same thickness.
- weld the panel in place. Make sure to alternate plug weld locations to control heat / panel warpage. When welding a lap joint, be sure to use a stitch or skip method to control heat and reduce panel warpage.



Refer to Module 1, “Demonstration: Vehicle Maker Welding Recommendations” in the presentation for an example of using collision repair information bulletins of different vehicle makers that show panel welding recommendations.



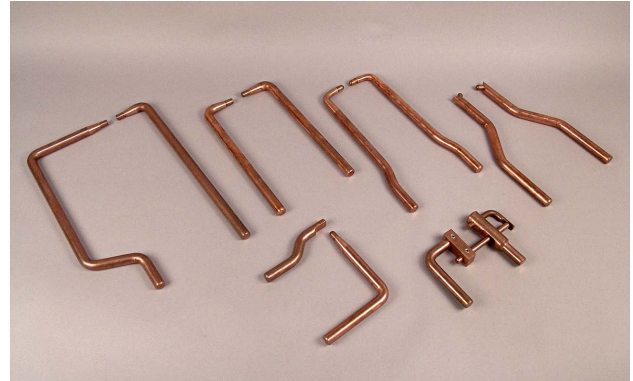
When ever possible, reassemble as the part was built from the vehicle maker.

Some vehicle makers recommend replacing vehicle maker spot welded panels with spot welds.

Some vehicle makers provide specifications for using spot welders which includes:

- distance between welds.
- recommended tip diameter.

- destructive test parameters.
- locations where GMA (MIG) welds are required.



These are just a few examples of the many different types of electrodes used for STRSW.

When using STRSW to attach panels:

- coatings should be removed from the spot weld area so tips contact bare steel.
- various arm configurations may be required depending on panel design. For example, the pinchweld around a door opening requires a different arm configuration compared to the pinchweld around the wheelhouse.

	Spot Welds	Plug Welds	Brazing
Honda	× ⊗ ⊠	●	⚡
Toyota	⊙	OMI	
Mazda	●	■	⊖⊖
Nissan	●	■	
Subaru	●	■	
Volkswagen	⊕	⊗	
Volvo	⊗	⊗	

Notice the different spot weld symbols used by the different vehicle makers.

When following vehicle maker recommendations for using STRSW, note that vehicle makers may use different spot weld symbols in their body repair manuals.



Plug weld holes are punched into the replacement part in accordance with the vehicle maker's recommendations.

GMA (MIG) plug welding requires:

- drilling or punching holes.
- identifying the number and location of plug welds as spot welds.
- a 5/16" (8 mm) hole if no recommendations exist.

Some vehicle makers provide recommendations for the number, locations, and size of plug welds required to attach a panel. Also, check the flange size. If the hole is too big for the flange width, a smaller plug weld hole is required.

	Spot Welds	Plug Welds	Brazing
Honda	× ⊗ ⊠	●	⚡
Toyota	⊙	OMI	
Mazda	●	■	⊖⊖
Nissan	●	■	
Subaru	●	■	
Volkswagen	⊕	⊗	
Volvo	⊗	⊗	

Notice the different plug weld symbols used by the various vehicle makers on this chart.

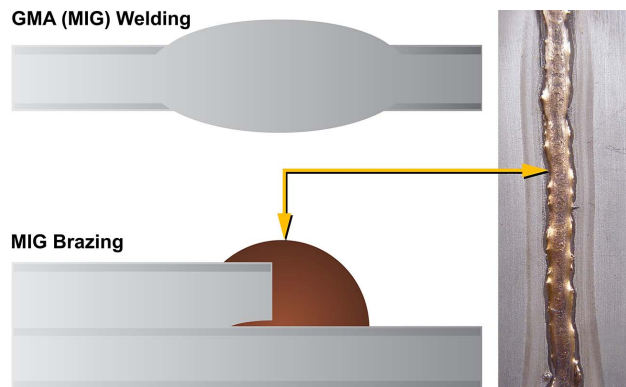
When following vehicle maker recommendations for using GMA (MIG) welds, note that vehicle makers may use different GMA (MIG) plug weld symbols in their body repair manuals.

When setting up a welder, a good starting point is one amp for every 0.001" of metal thickness. For example, 18 gauge steel requires 48 amps and 14 gauge steel requires 75 amps. Keep in mind, the welder may need to be adjusted for different material thicknesses to achieve a good weld.

Types of electrode wire commonly used include:

- AWS ER70S-6.
- AWS ER70S-7.

Electrode wire thickness is either .023 or .030 for exterior panels.

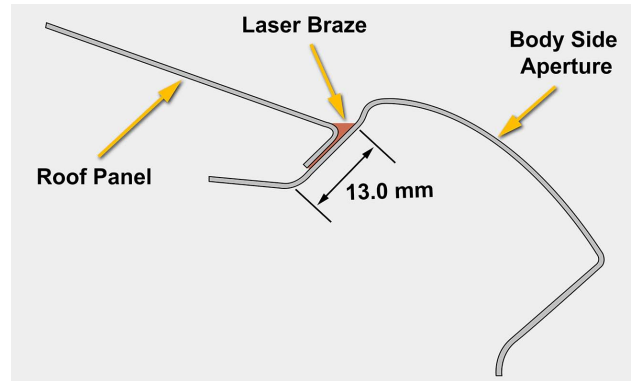


In this image, the MIG brazing material is melted and flows into the joint through capillary action.

Replacing at factory seams may include locating silicon bronze joints on the panel, this material may look like brass. Silicon bronze is applied using a GMA (MIG) process; this is commonly referred to as MIG Brazing. MIG brazing produces a small heat affect zone.

There are different alloys of electrode wire for this process therefore it is important to reference service information for correct selection.

This should not be confused with oxyacetylene torch brazing, which is not commonly recommended due to the large heat affect zone. It is important to note that neither process is designed to fuse the joint but forms an adhesive bond with the base metal.



This crosscut diagram shows an example of silicon bronze applied at the factory between the roof panel and the body side aperture.

Common areas for using silicon bronze or brass include the:

- joint between the roof and quarter panel.
- joint between the roof and A-pillar.
- tail lamp area.
- joint between the rear body panel and quarter panel.



The vehicle maker's attachment procedure will specify the type of adhesive required.

Adhesive materials used generally include a two-part epoxy unless otherwise noted by the vehicle maker. Adhesive is to be

applied following the adhesive maker's recommendations and are generally applied over:

- a bare steel flange.
- an E-coated or primed flange.
- a layer of existing adhesive (generally for urethane applications only).

Common adhesive types include:

- acrylic, which bonds all types of bare metals and are excellent for cross bonding aluminum to steel. They have good NVH properties and some offer anti-corrosion properties, so primers on bare metals may not be necessary. Most acrylics have a fast room temperature cure and are stable with regards to temperature and moisture during cure.
- epoxy, which bonds well to ridged and semi-ridged plastics and are generally easy to sand and feather edge. Some may be too ridged for flexible substrates and they often require primers on bare metal applications. Epoxies can be heat cured to increase strength and accelerate the curing process.
- urethane, which is typically flexible and bonds well to plastics. However, they usually require primers on metal surfaces to protect against corrosion. Urethanes have good seam sealing and NVH qualities and are frequently the optimal choice for seam sealers. They are sensitive to

moisture during cure, packaging and storage.

Be sure to follow each adhesive maker's recommendations when bonding panels.



A hammer and dolly can be used to form a hem flange.



A hem flange tool forms a uniform, rounded hem flange.



A pneumatic hammer and dolly tool produces a well formed hem flange similar to a manual hammer and dolly.



The power drill hammer and dolly tool uses a standard power drill to operate the hammer and dolly function.

When installing a panel using a hem flange:

- apply adhesive to the panel where the hem flange will be made. Adhesives should be applied to both sides of the flange to seal out moisture and contaminants. If a door panel is being replaced, adhesive should be applied to both sides of the door frame flange that the door panel will be attached to.
- form a hem flange using a:
 - hammer and dolly

- hem flange tool
- pneumatic hammer and dolly tool
- power drill hammer and dolly tool
- remove excessive adhesive from the panels.

General Panel Finishing Procedures



Corrosion protection is being applied to the backside of the panel.

To finish exterior panel installation:

- dress welds as required. Do not thin adjacent metal.
- apply silicon bronze / brass in the joint area, if required.
- apply body filler. Do not apply body filler to stationary glass pinchweld flanges.
- restore corrosion protection. This should be done on both the interior (with a spray wand if necessary) and exterior surfaces.



A foam pad sound deadener is applied to the replacement part before it is installed on the vehicle.

Transfer sound deadening materials from the vehicle maker panel to the replacement panel if possible or install new sound deadening material if necessary.

Types of sound deadening materials include:

- foam pads.
- foam blocks.
- foam filler.
- anti-flutter material.

Sound deadening materials are:

- loosened from the panel with heat.
- generally replaced with a new piece.

Personal Safety

A combination particulate / vapor respirator should be worn when installing or heating some sound deadening materials.

Module Wrap Up

Topics discussed in this module included:

- types of panel construction.
- measuring for damage.
- making repair or replace decisions.
- what to include in a repair plan.
- understanding the function of silicon bronze and brazing.
- removing and installing spot welds and GMA (MIG) welds.

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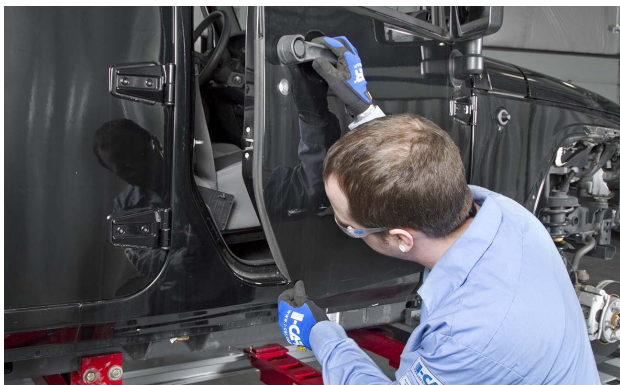
Module 2 - Door Skin Replacement

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Door Damage Analysis

Learning objectives for this module include:

- making damage analysis decisions for door skins and door shells.
- creating a repair plan for door repair.
- removing a door skin from a door shell.
- installing a door skin.



The door is opened and closed during initial inspection to make sure it works properly.

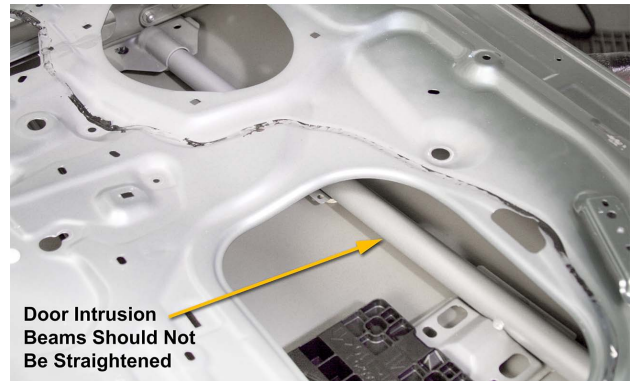
Before removing the door, make sure that the:

- vehicle structure has been restored to specification.
- door shell fits properly.
- door opens and closes properly.

Improper door shell fit may result in:

- water leaks.
- wind noise.
- rattles.

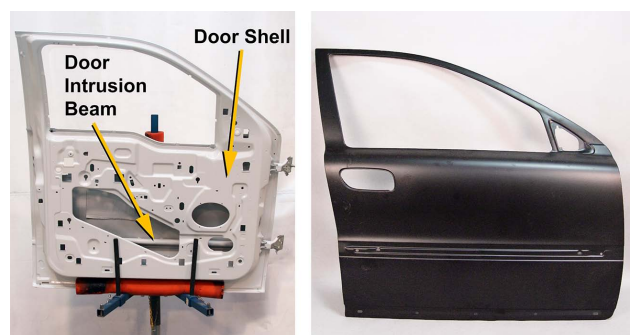
- improper window operation.



Intrusion beams should be inspected for damage.

Intrusion beams are important to the occupant as they provide protection during a side-impact collision. Intrusion beams are typically made of ultra-high-strength steel. If the intrusion beams are damaged in any way, they must be replaced. There are no repair recommendations. Most require replacing the door shell.

Some vehicle makers provide a replacement part and allow intrusion beam replacement. The replacement part is bolt-on only.



Damage to the intrusion beam and door shell are considered when determining if the complete door shell needs to be replaced.

The decision to replace the door shell is a subjective decision based on:

- the extent of damage.
- damage to the door shell structure.

Planning A Door Repair



This door skin requires adhesive bonding during the installation process.

When developing a repair plan, determine:

- if the door shell or intrusion beam is damaged.
- the vehicle maker's door skin part availability for replacement.
- if replacement or sectioning procedures are available from the vehicle maker.
- where the door skin should be bonded and / or welded.



This door skin is hem flanged by the vehicle maker.

Door skin bonding is an approved repair method by many vehicle makers and is the most common method of door skin replacement.

When using adhesives to attach the door skin, the skin will be hemmed over the inner flange, sandwiching the adhesive and door frame between the two panels.



Adhesive is applied along the outer flange on this door shell.

Repairs may require combining welding and bonding. Adhesives may be used along the outside flanges while welds are used in the pinchweld of the window opening or a sectioning joint.

This should not be confused with the process of weld-bonding, which uses adhesive and spot welds in the same joint.



In preparation for a door skin repair, this technician is removing the interior molding and trim.

Door trim and glass removal is recommended to protect door parts from damage during door skin replacement. With the door on or off the vehicle, remove the:

- door trim.
- door glass, and disassemble run channels.
- door handle and door lock.
- seat belt assemblies in the door panel (if equipped).
- airbags and sensors.

It is important to properly store and organize all removed fasteners during the disassembly process.

Door removal requires disarming side airbags (if equipped). Also, disconnect wiring, being sure to mark the wire routes and wire locations with numbered tape. Numbered tape can also be used to mark

any fastener locations. Also the practice of using photos will help the reassembly process.

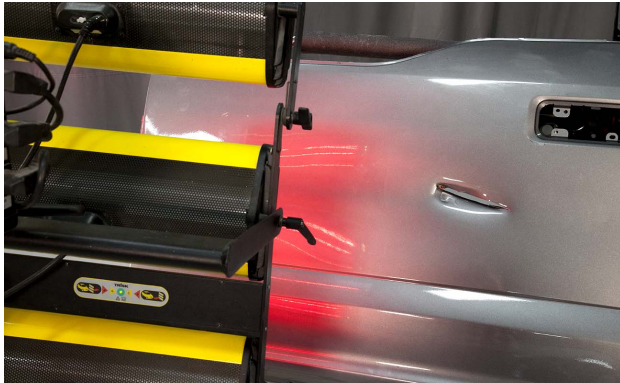
Removing A Door Skin



A grinder is used to remove the hem flange edge.

One method of removing a door skin includes:

- removing the edge of the hem flange using a disc grinder. Grind until separation between the outer panel and the door shell is visible. Some panel designs may require cutting an access hole to help remove the skin.
- removing the spot welds around the window frame (if equipped).



An infrared lamp is positioned near the bonded flange to soften the adhesive.

One method of removing a door skin includes:

- separating the bonded flange (if bonded) using heat not to exceed 400°F.
- removing the remaining strip of flange and the door panel with a hammer and chisel (if necessary). Do not damage the door frame.

Installing A Door Skin

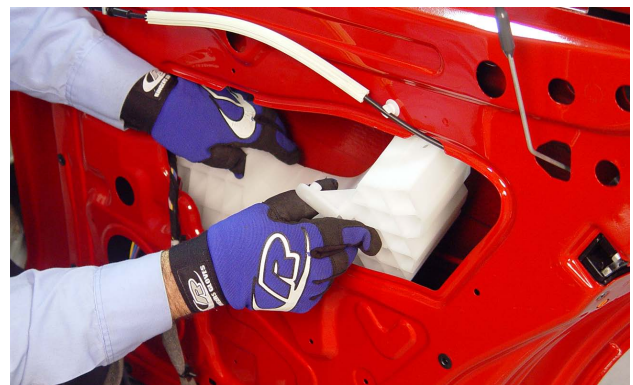


A plastic-coated abrasive wheel is used to remove the remaining adhesive from the flange.

Remove any remaining adhesive from the flange area. A plastic-coated abrasive wheel is recommended to prevent

thinning the metal. Remove any damaged foam or sealant from the intrusion beam.

Align and straighten the door shell flange using a hammer and dolly. Be sure to restore the shape and contours of the door shell. If the door shell is not properly repaired, it is possible that the door skin will not fit properly when it is fit up against the door shell.



This technician is installing silencer pads inside the door shell.

Perform a test fit of the replacement skin on the door shell. Temporarily clamp the shell in place and install the door on the vehicle. Using locking pliers to hold the panel in place during the fit test will prevent the door from closing. Therefore, hem a small area of the flange on both sides of the door skin to hold the panel in place.

Make sure the vehicle is resting on its suspension. Check the panel gaps. If the door fits, install screws in existing holes to secure the panel in place (if there are existing holes available). These screws will also be used to realign the panel

once it has been removed to apply the adhesive.

Remove the door and door skin, and install silencer pads (if equipped) on the skin backside.



With the many different types of adhesives available, always follow the vehicle maker's recommendations.

Follow adhesive maker's recommendations for part preparation and product application. Some adhesives:

- may require application over the E-coat.
- contain anti-corrosive properties.
- may require application over bare steel. This requires removing E-coat, primers, or paint coatings, from the flange on both the door skin and door shell.



An adhesive bead is applied along the flange edge.

Apply adhesive to the flange and to the intrusion beam. Some intrusion beams have anti-flutter foam or sealer between the door shell and skin. It is critical to use the vehicle makers recommended material. If the flange has been stripped of E-coat, make sure the adhesive is spread over the entire flange to ensure all bare metal is covered.

Repeat this process on the door skin. To ensure proper bond-line thickness, a third bead may be applied to either the door skin or door shell. This bead does not get spread out.



A hem flange tool is used to form a hem flange on this door skin.

Install the door skin onto the door shell. Slide the panel when adjusting it, do not

lift the panel. Reinstall temporary fasteners (if used) and clamp the panel in place to hem the flange. The flange can be hemmed using a:

- body hammer and shot dolly. The shot dolly is used to prevent damage to the skin.
- specifically designed hemming tool. This particular tool is used with a ratchet and designed to fold over the hem flange edges and crimp it to the door shell.

Install the door on the vehicle and check operation and panel gaps. Ensure the door fits the vehicle before the adhesive cures. Adjust accordingly during the adhesive cure time, remove the door and complete hemming the flange. Apply corrosion protection and seam sealers where necessary.



Refer to Video: Door Skin Replacement in the presentation. This video discusses concepts on how to remove and install a door skin.

Module Wrap Up

Topics discussed in this module included:

- making damage analysis decisions for door skins and door shells.
- creating a repair plan for door repair.
- removing a door skin from a door shell.
- installing a door skin.

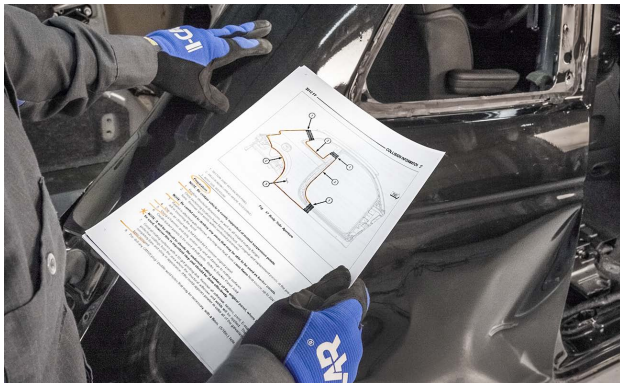
***Module 3 - Quarter
Panel, Box Side, Cab
Corner***

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Planning A Quarter Panel Repair

Learning objectives for this module include:

- developing a repair plan.
- identifying procedures required for removing and installing a complete quarter panel.
- identifying procedures required for removing and installing a sectioned quarter panel.
- identifying procedures required for removing and installing a truck box side.
- identifying procedures required for removing and installing a truck cab corner.
- identifying procedures required for sectioning a truck cab corner.



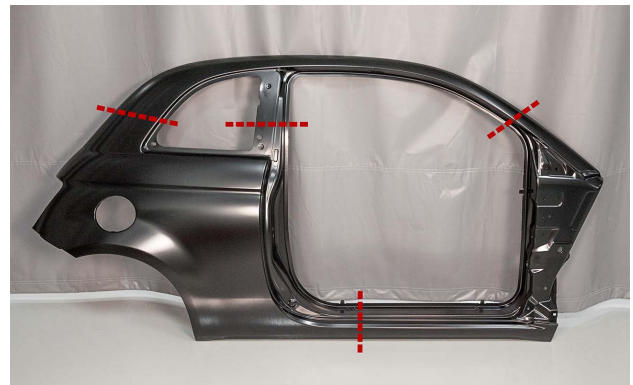
Published procedures from the vehicle maker are being used to consider possible sectioning locations.

Determine if the panel can be sectioned or if it should be replaced at factory seams.

Considerations include:

- location and extent of damage.
- published procedures from the vehicle maker.
- does the panel design allow it?

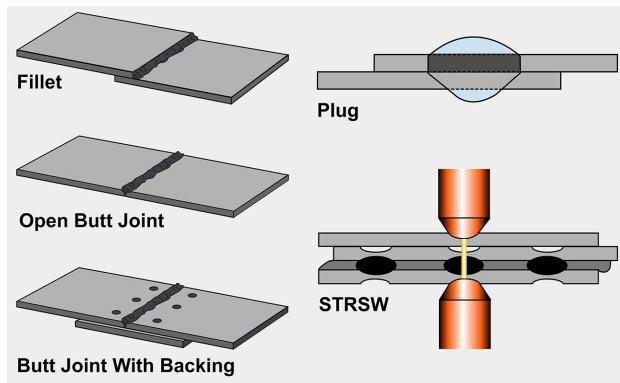
Replacing a quarter panel at the factory seams is recommended by several vehicle makers. However, this may require replacing the roof panel depending on flange design.



Some common vehicle maker recommended sectioning locations are shown.

Follow vehicle maker recommendations for quarter panel sectioning. Common sectioning locations include:

- through the quarter glass or sail panel area (most common).
- at the belt line.
- at the dog leg.

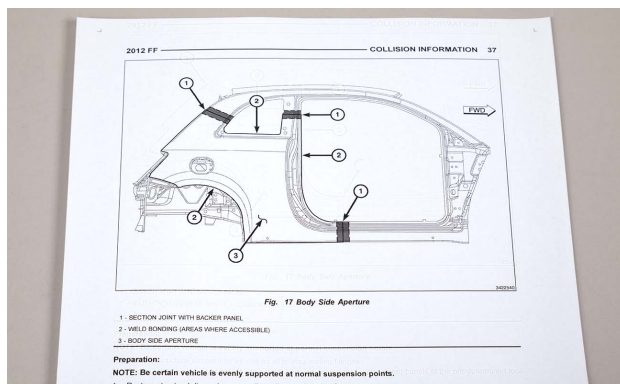


These are examples of the most common welds used for a quarter panel-type repair.

Quarter panel replacement requires GMA (MIG) welds that include:

- fillet joint.
- open butt joint.
- butt joint with backing.
- plug.

Spot welds are used along mating flanges / pinchwelds. GMA (MIG) welds are used where there is no access to the backside of the weld.



Weld bonding locations may be specified by this vehicle maker's body repair manual guidelines.

Weld bonding quarter panels is recommended by some vehicle makers. Weld bonding is the process of combining

adhesives with spot welds or GMA (MIG) welds. Refer to the body repair manual for guidelines on location of adhesive and welds.

If weld bonding using MIG welding, it is important to ensure that there are no adhesives in the weld area.

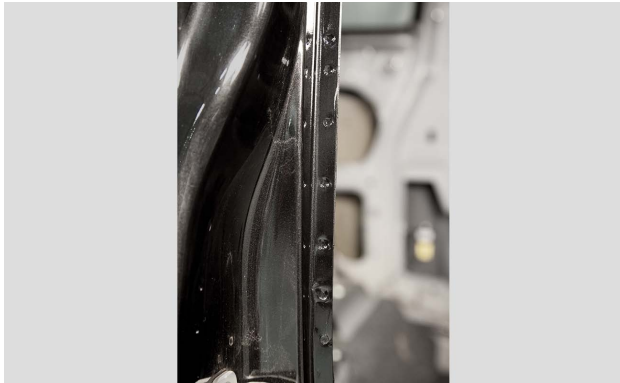
Removing A Quarter Panel



This vehicle is being prepped for a quarter panel repair.

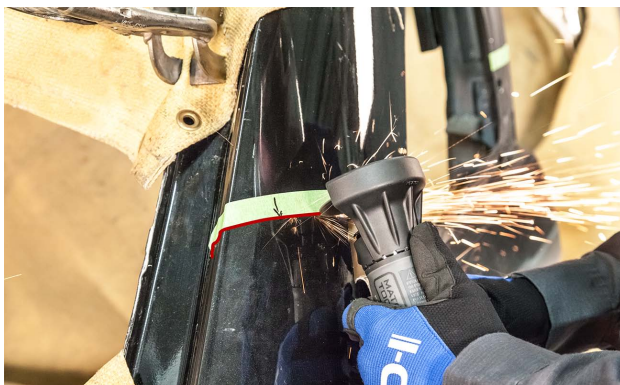
Quarter panel replacement may require the removal of the:

- backglass.
- quarter glass.
- fuel filler tube.
- wiring on the panel backside.
- deck lid.
- tail lamp assembly.
- bumper cover.
- SRS and airbags.
- interior trim.



Spot weld locations have been identified for removal.

To remove a quarter panel, make sure the vehicle structure and adjacent panels are in alignment before the quarter panel is removed. Identify and mark spot weld locations.



A lap and scribe joint method is being used where the first cut slightly longer than the final cut location.

Determine the quarter panel cut location. If a cut must be made in the sail panel area, make the first cut slightly longer than the final cut location.

At the rough cut locations where reinforcements are suspected, the technician should cut a window to see if there are any reinforcements underneath the outer panel. Caution will need to

be taken so that we don't damage the reinforcement.

If the joint is in the roof panel, be sure to only remove the spot welds along the quarter panel flange. Brass or silicon bronze in the factory seam along the roof rail may need to be ground out to separate the roof panel and quarter panel. Do not damage parts attached to the quarter panel which are not to be replaced.



A grinder is being used to remove spot welds.



The sectioned quarter panel is removed after all the welds and adhesive bonds have been removed.

Remove the spot welds along all the remaining flanges, wheelhouse, rear body panel, quarter glass, etc.

Use heat to loosen NVH materials. These are typically located along the wheelhouse or fuel filler area.

Remove the quarter panel.



Refer to Video: Quarter Panel Removal in the presentation. This video discusses concepts of removing a quarter panel by sectioning the panel.

Installing A Quarter Panel



The mating surface of this replacement part is being prepared for adhesive.

To install a complete quarter panel, prepare the vehicle flanges by removing burrs and spot weld nuggets, straightening the flange, etc. Clean the mating surfaces.



The replacement panel is clamped to the vehicle to ensure that all sectioning joints fit prior to welding and bonding.

Test-fit the replacement panel and clamp it in place. Make measurements from various points on the quarter panel. If no measurements exist from the vehicle maker or measuring equipment maker, comparative measurements will be required.

Install fasteners to help realign panels after panel removal.

Determine if STRSW or GMA (MIG) welds should be used. Also, if it has been determined that plug welds will be used, and there are no recommendations for plug weld hole size, use a 5/16" hole, contingent on the flange size supporting a hole of this size. Make sure they are in the same weld locations used originally by the vehicle maker.

Remove the panel from the vehicle and apply weld-through primer to all bare metal areas if recommended by the vehicle maker.



Adhesives are spread using an acid brush to cover all bare metal areas.

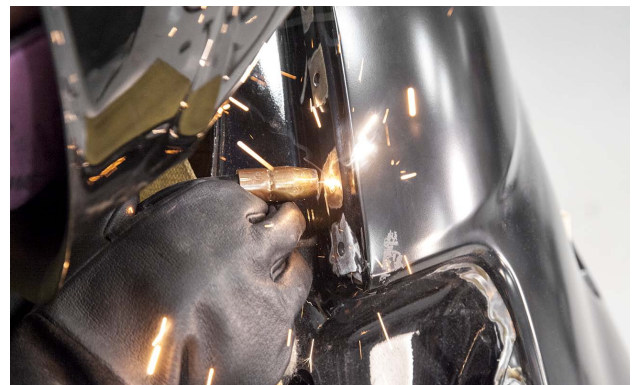
Typical adhesive application requires:

- applying a bead of structural adhesive to the area where the two panels are to be bonded and weld bonded.
- smoothing the adhesive with a body filler spreader, acid bush, or equivalent to cover all bare metal surfaces.
- applying a second bead of adhesive. This bead is not smoothed out to ensure proper adhesive thickness.

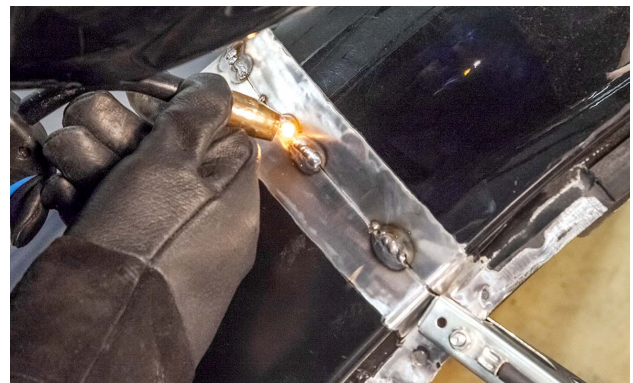
Once the panel is set in place, if the panel needs to be adjusted, it should be slid and not lifted. Lifting will cause air bubbles and will weaken the bond.



Spot welds are made along the window flange.



Some exterior panel installations require plug welds.



This technician is performing a skip weld technique to make this butt joint with backing sectioning joint.

Position the quarter panel on the vehicle and clamp it in place.

Recheck alignment using adjacent panels.
Check the panel fit with the suspension

loaded to ensure that sag is not affecting proper panel alignment.

Make test welds, clean the weld zone so it is free of weld-through primer, then make the required welds.

If required, replace:

- sound deadening material.
- mesh resin material at the sail panel for reinforcing.
- stiffeners at the wheelhouse to sail panel.



Corrosion protection is being sprayed inside the rocker panel after the repair is complete.

Dress the welded area and finish the joint areas if required. Apply corrosion protection, primer, seam sealers, etc.

Complete cosmetic repairs. Reinstall glass, interior, fuel tanks, etc.



Refer to Video: Quarter Panel Installation in the presentation. This video discusses the different welding and bonding installation techniques used for installing a sectioned quarter panel.

Removing And Installing Truck Box Sides



A grinding wheel is used to remove the exposed spot welds on this truck box side.

Truck box sides are either steel, aluminum, or plastic that are either bonded, fastened, or welded together.

To remove a welded box side:

- remove the box from the truck.
- remove any necessary trim and related parts such as tailgate, tail lamps, etc.
- locate and remove spot welds.

- remove the side panel.



In preparation for installation, the technician is determining the types of welds needed.

To install a truck box side:

- determine if STRSW or GMA (MIG) welding will be used.
- attach the panel in place and check fit and gaps. Make sure box contours align with cab contours. This may require temporarily installing the box.
- remove the box.

If the box side is being installed with the box off the vehicle, care must be used to ensure correct alignment. Ensure that the bed is on a level platform that does not allow the bed to twist.

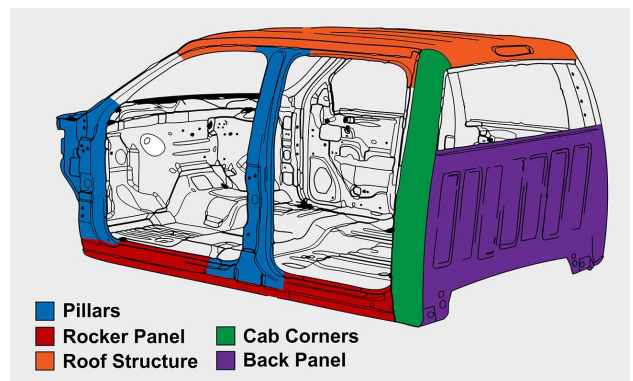


All trim, including this tail lamp, are installed when completing the repair.

To install a truck box side:

- drill or punch plug weld holes, if necessary.
- make test welds, then weld the panel in place.
- dress the welds and prep for refinish.
- prime panel in weld areas and refinish the box side.
- apply corrosion protection after refinishing.
- install on vehicle and trim out.

Removing And Installing Cab Corner Panels



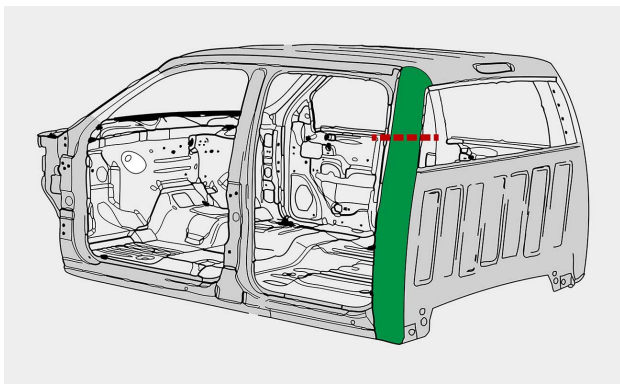
The vehicle makers panel design plays a key part in determining the repair plan on this extended cab.

Vehicle maker joints from the quarter panel / cab corner to the roof panel are typically a lap joint spot welded along the flange. Some vehicle makers may install silicone bronze to hide the seam. The rear cab panel to cab corner is also spot welded in place along the flange.

Determine if the part can be sectioned or if it must be replaced at factory seams. Similar to sectioning quarter panels, this decision is based on:

- extent and location of damage.
- panel design.
- vehicle maker recommendations.

Also similar to quarter panels, the roof-to-cab corner joint may require sectioning to avoid removing adjacent panels.



This shows a typical sectioning joint location on a extended pickup.

Sectioning joints are typically made about 3/4 up the cab corner. If it is an extended cab, joints are commonly made in the window area. Sectioning joints are either open butt joints or butt joints with backing.

Module Wrap Up

Topics discussed in this module included:

- developing a repair plan.
- procedures required for removing and installing a complete quarter panel.
- procedures required for removing and installing a sectioned quarter panel.
- procedures required for removing and installing a truck box side.
- procedures required for removing and installing a truck cab corner.
- procedures required for sectioning a truck cab corner.

Module 4 - Roof Panel Replacement

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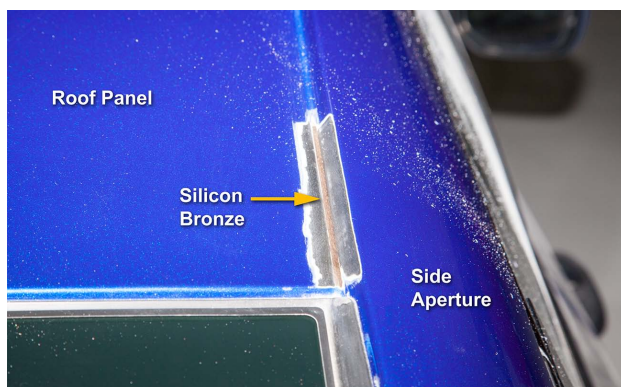
Roof Panel Designs

Learning objectives for this module include:

- identifying roof panel joint designs.
- identifying types of vehicle maker attachment methods.
- removing and installing welded-on roof panels.
- removing and installing bonded-on roof panels.



This roof drip channel is covered by a piece of black trim molding.



The silicon bronze weldment is revealed after the coatings are removed.

Roof panels that have a channel between the roof and quarter panel use either a

trim piece or seam sealer to cover the seam. Panels where the roof and quarter panel appear as one piece use silicon bronze (or brazing) to cover the seam.

Some roof panel flange designs require sectioning the quarter panel to replace the roof. In other situations, the installation of a uniside may require the roof panel removal and replacement or partial removal of the roof section.



This sunroof design is integrated into the roof panel and may require removal and installation for roof panel removal.

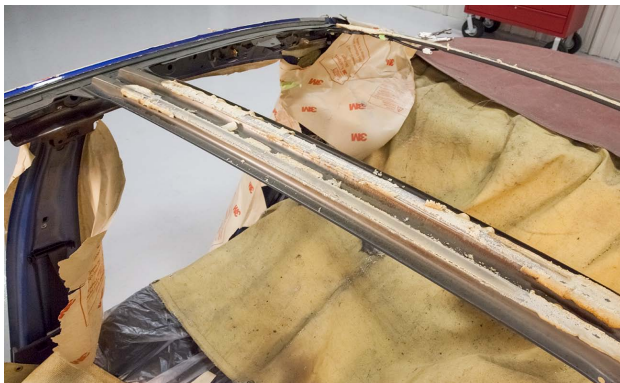
Roof panels may be equipped with a sunroof, which is typically removed and reinstalled into the new roof panel following replacement. Sunroofs are fastened in place. Follow the vehicle maker's procedure for removal and installation.

Also, note if the sunroof is aftermarket. If the sunroof is damaged and must be replaced, it may require ordering from an aftermarket vendor compared to ordering from the vehicle maker.



The roof on this vehicle was attached with a laser brazed joint on the sides, spot welds at the headers, and anti-flutter material on the roof bows.

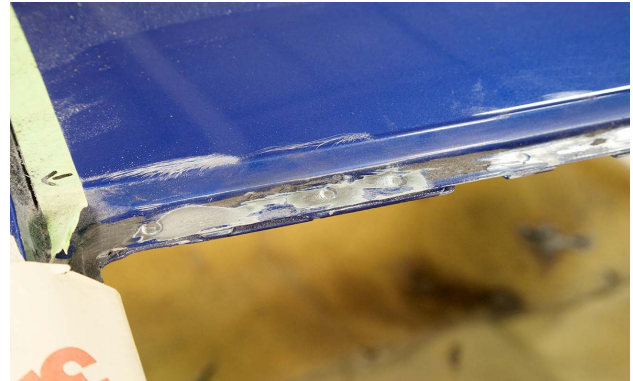
Roofs may be connected at the A-pillar, B-pillar, and / or the sail panel area. The joint design could have multiple layers. Refer back to the discussion on quarter panel-to-roof joint designs.



With the roof panel removed, the anti-flutter material is exposed on the roof bows.

Vehicle makers place anti-flutter materials between the roof bows and the roof panel. Anti-flutter materials prevent the roof panel from moving up and down due to air movement or pressure. Sound deadeners are also placed on the panel to reduce outside noise.

Panel Attachment Methods

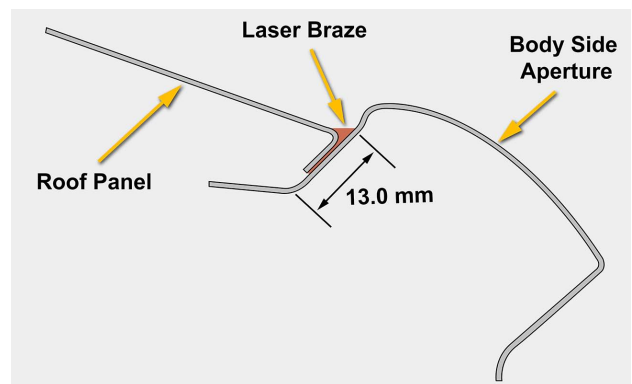


The vehicle maker has welded the roof panel front and rear headers using spot welds.

The roof panel is:

- welded and / or bonded to the front and rear headers / roof bows.
- typically bonded to the roof bow between the front and rear headers.

Planning A Roof Panel Repair

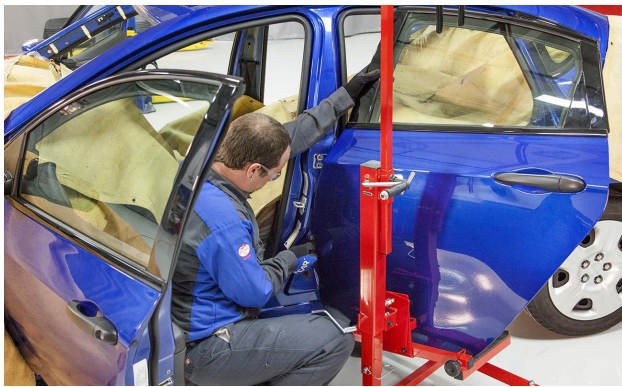


Using a vehicle makers diagram will help determine the repair sequence.

When replacing a roof panel, determine:

- if this requires additional panels to be replaced.
- the replacement sequence. This may be determined by the roof flange assembly.
- all electrical parts to be removed. Electrical parts may include dome lamps, stereo control, heat or air conditioning controls, etc.

Glass removal is generally required for roof panel installation. Both the windshield and backglass are typically removed. Some designs, however, allow the glass to remain installed during roof panel replacement.



The doors are removed to protect the exterior and interior finish surface as well as provide easier access for the technician.



Moldings are removed to prevent damaging them during the repair process.



Side airbags are removed prior to repair to protect the technician.

Roof panel replacement generally requires removing:

- interior parts such as seats and carpeting.
- headliner.
- doors.
- heating / cooling tunnels.
- windshield and backglass.
- sunroof drain panels.
- antennas.
- luggage racks.

Additional items that must be removed or disassembled before roof removal include:

- sunroofs (if equipped).
- electrical accessories.
- moldings.
- drain tubes.
- airbags or side airbag curtains (if equipped).

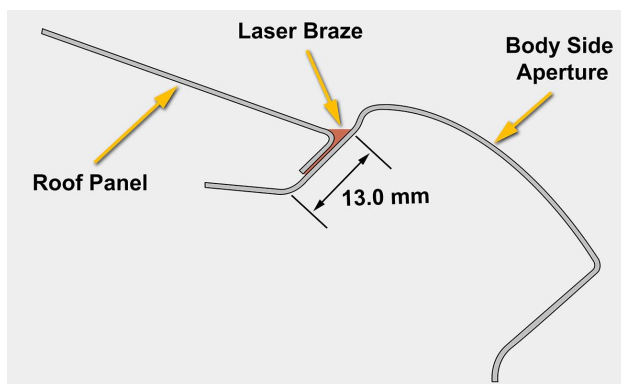


Once the roof panel is removed, inspect the roof bows for damage.

Roof bows:

- provide support for the roof. The number of roof bows vary from vehicle to vehicle.
- are welded into place when replaced.
- that are damaged need to be repaired or replaced depending on the severity of damage.

Roof bows have an adhesive and / or anti-flutter material between it and the roof panel. Heat may be required to loosen during removal using a utility knife, piano wire, or pneumatic cutting tool.



This roof panel is placed on the inside of the body side aperture.

If replacing multiple panels, it is important to determine the proper replacement sequence. To do this, determine where the roof panel is placed in relation to the pillars and sail panel-between, on top, or beneath.

Removing And Installing Weld Bonded Roof Panels



This technician is using a grinding wheel to cut the laser brazed weld.

To remove a weld bonded roof panel:

- make sure adjacent panels and the vehicle structure are in alignment before removing the roof panel.
- some vehicle makers recommend using a cutoff wheel or reciprocating saw to cut a center section out of the roof panel for access. Do not cut into the roof bows.
- separate welds along the roof edges.



A drill is used to remove the spot welds in the header section of the roof panel.



Heat can be used to soften adhesives or anti-flutter material on the roof bows.



With the laser brazing, spot welds, and adhesives removed, the roof can be separated from the vehicle.

To remove a weld bonded roof panel:

- identify spot weld locations and remove remaining spot welds.

- release any adhesive using heat from a no-flame heat source.
- remove the roof panel and straighten vehicle flanges.



To remove the remaining flange, grab the flange with a pliers and bend it back and forth until it releases from the weld.



The remaining weldment is ground off using caution to not remove any base metal.

To remove a weld bonded roof panel:

- remove remaining roof flange. This flange can be removed by grinding off or by grasping with a pliers and moving back and forth until it breaks free of the weld. This will be repeated for the entire length of the flange.

- grind off remaining weld. Use caution when grinding to only remove the weld material and not the base metal.



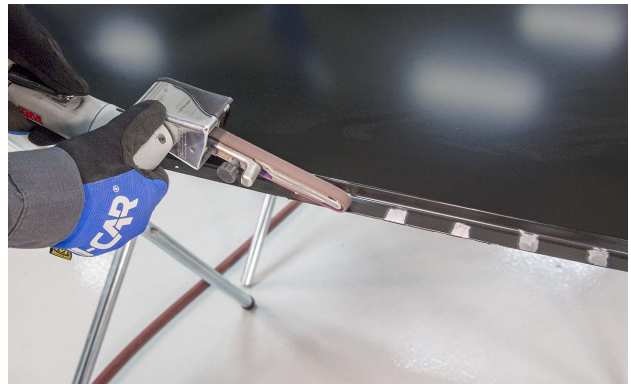
Refer to the Video: Removing A Weld Bonded Roof Panel in the presentation. This video demonstrates the procedure and skills required for removing a weld bonded roof panel.



In preparation for weld bonding, the headers are cleaned to bare metal.

To install a weld bonded roof panel:

- perform a test-fit of the roof panel.
- clean the mating surface on the vehicle headers and replacement part.



Remove the coatings only where the spot weld locations have been marked.

To install a weld bonded roof panel:

- remove coatings from the replacement panel.
- determine if the panel should be welded using spot welds or GMA (MIG) plug welds.
- drill plug weld holes where required. Punch or drill 5/16" holes in the replacement panel if no vehicle maker recommendations exist. If using a lap joint, allow for a minimum of 1/4" overlap. If using spot welds, refer to the vehicle maker's recommendation for the electrode diameter, weld locations, and spacing.



The vehicle maker procedure recommends that the front header flange is held in place with adhesive and spot welds.

To install a weld bonded roof panel:

- position the roof panel on the vehicle.
- check fit and gaps.
- make test welds.
- apply adhesive.
- weld the panel in place.



Refer to the Video: Installing A Weld Bonded Roof Panel in the presentation. This video demonstrates the procedure and skills required for installing a weld bonded roof panel.

Removing And Installing Bonded-On Roof Panels



This technician is using a pneumatic knife to cut the urethane adhesive along the roof bow.

If removing a roof panel that is bonded on with urethane:

- cut an access hole if necessary.
- unscrew fasteners if equipped.
- use piano wire or a pneumatic knife to cut adhesive.
- remove the roof panel.
- trim the remaining adhesive to ensure a smooth surface for the replacement adhesive.

Vehicle Protection

- Do not gouge the flanges or roof bows with the scraper.
- Replace all damaged corrosion protection.



This roof has urethane applied according to the vehicle maker's instructions and is clamped in place while curing.

To install a bonded-on roof panel:

- clean the area to be bonded on the roof panel and the vehicle. Apply corrosion protection where necessary.
- prepare the bonding area per the adhesive maker's instructions.
- apply the adhesive.
- clamp in place or install fasteners or GMA (MIG) plug welds if required by the adhesive maker.

Roof Installation - Finishing



During the weld bonding process, adhesive is squeezing out and will need to be removed.

Complete the repair by:

- dressing the welds, if required.
- removing adhesive squeeze-out (if bonded).
- applying corrosion-resistant primer, seam sealers, etc. as required to duplicate appearance.
- refinishing as required.
- reinstalling all trim, wiring, glass, etc. that were removed.

Module Wrap Up

Topics discussed in this module included:

- roof panel joint designs
- types of vehicle maker attachment methods.
- removing and installing welded-on roof panels.
- removing and installing bonded-on roof panels.

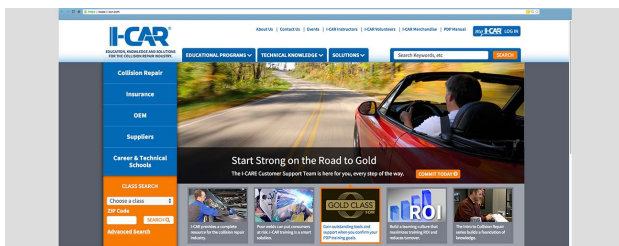
Module 5 - Rear Body Panel Replacement

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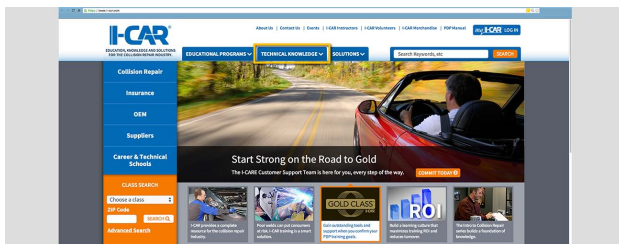
OEM Rear Body Panel Remove And Install Procedures

Learning objectives for this module include:

- determining where to locate OEM rear body panel removal and installation procedures.
- examining procedures for removing and installing rear body panels.



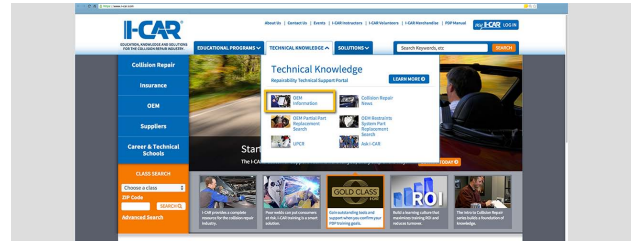
The i-car.com website is a useful and easy launching portal to find the most up-to-date vehicle maker's collision repair manuals.



The I-CAR home page has many easy to access tabs and links to help the collision repair industry find the information they are looking for.

The tabs across the top include Educational Programs / Technical

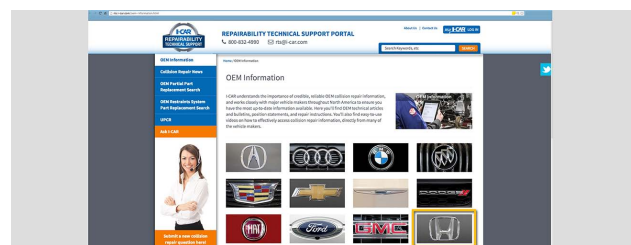
Knowledge / Solutions. To find the most up-to-date vehicle maker body repair manuals, select the **Technical Knowledge** tab.



The **Technical Knowledge** tab is the link to the Repairability Technical Support (RTS) Portal. Here the user can find:

- OEM Information.
- OEM Partial Part Replacement Search.
- Uniform Procedures For Collision Repair (UPCR).
- Collision Repair News.
- OEM Restraints System Part Replacement.
- Ask I-CAR.

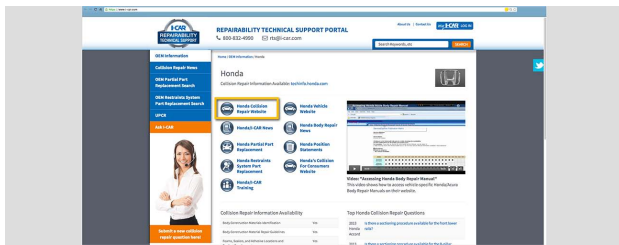
To find the most up-to-date vehicle maker body repair manuals, select the **OEM Information** link.



The **OEM Information** link directs the user to major vehicle maker's repair sites throughout North America. Here the user can find:

- OEM technical articles.
- OEM bulletins.
- position statements.
- repair instructions.

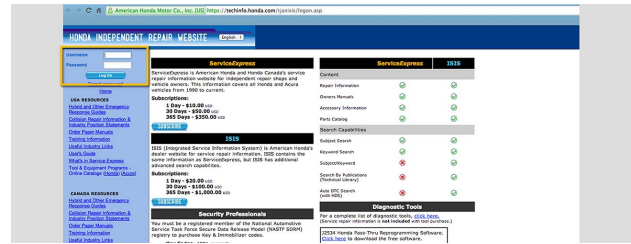
Select the OEM emblem for access to that OEM's information link.



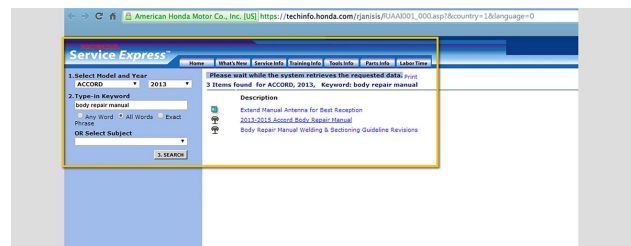
The vehicle maker's specific OEM information page is the users "one-stop-shop" for everything vehicle maker specific. On this page, the user can find:

- a direct link to the vehicle maker's collision repair website.
- vehicle maker news.
- partial parts replacement recommendations.
- restraint system part replacement recommendations.
- vehicle maker training.
- vehicle maker consumer website.
- body repair news.
- position statements.
- collision for consumers website.

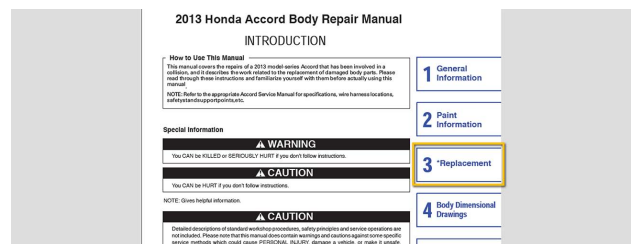
Each vehicle maker will have similar information available on their specific OEM information page.



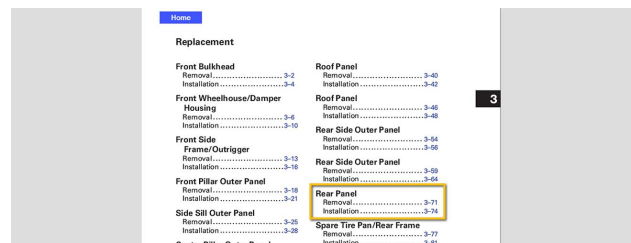
Depending on the vehicle maker, the repair website may be free or require a subscription to access.



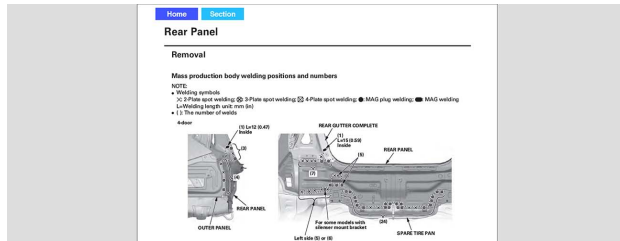
Vehicle makers require the user to select the model and year of the vehicle's body repair manual they are requesting.



Vehicle makers body repair manuals may have hyperlinks within the document to quickly access the specific sections of the manual.



This vehicle maker's body repair replacement pages are hyperlinked to quickly access the specific rear panel removal and installation sections of the manual.



Refer to the Activity: Examine Rear Panel R&I Procedure in the presentation. This activity will have the student, using a vehicle maker's rear body panel removal and installation procedure, examine the steps as they relate to previously taught exterior panel removal and installation steps noting the similarities.

Module Wrap Up

Topics discussed in this module included:

- locating OEM rear body panel removal and installation procedures.
- examining procedures for removing and installing rear body panels.

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